EPFL

Multi-Agent Reinforcement Learning: Theoretical Advances in Sample-Efficient Learning

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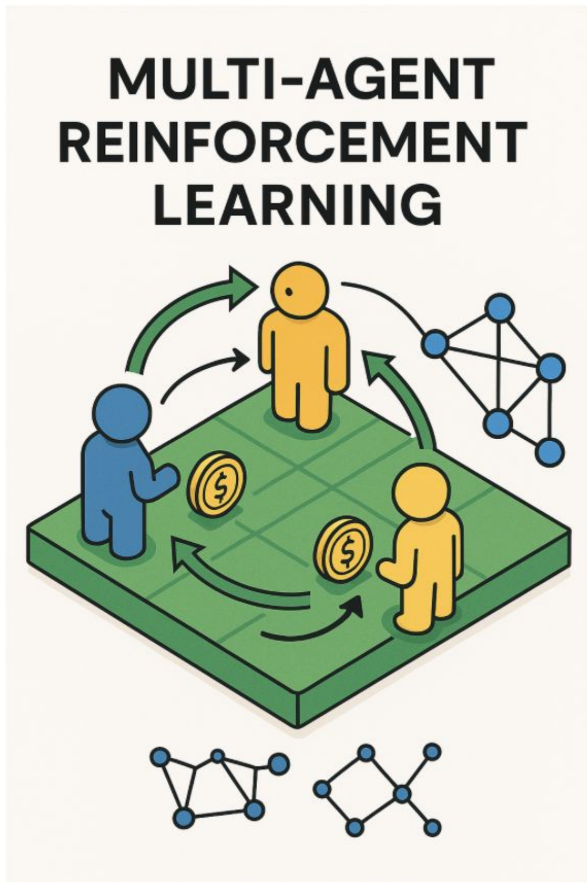
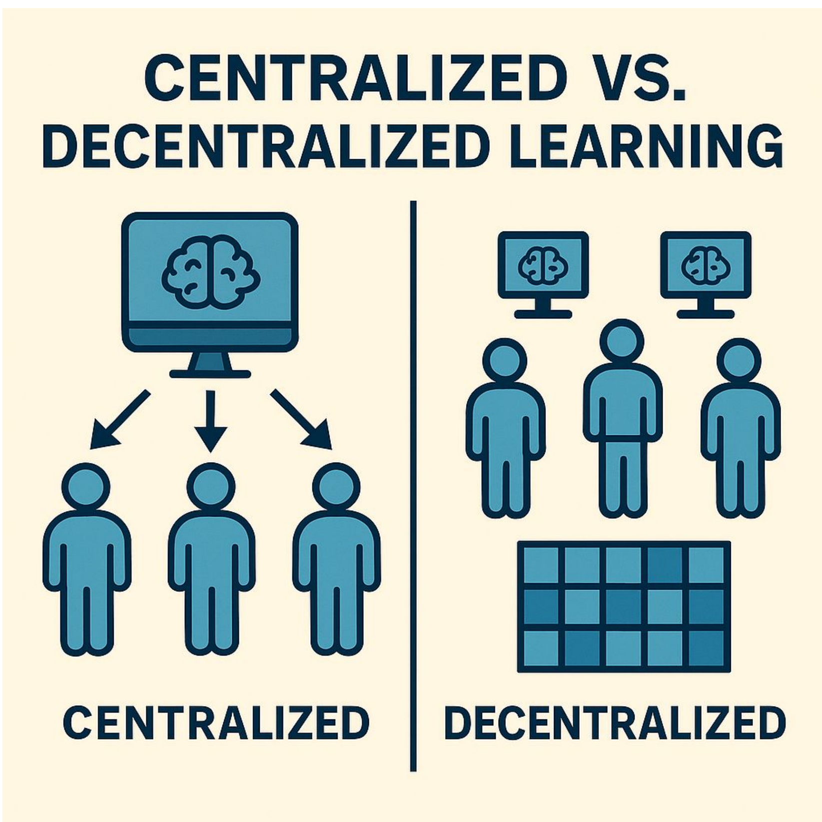
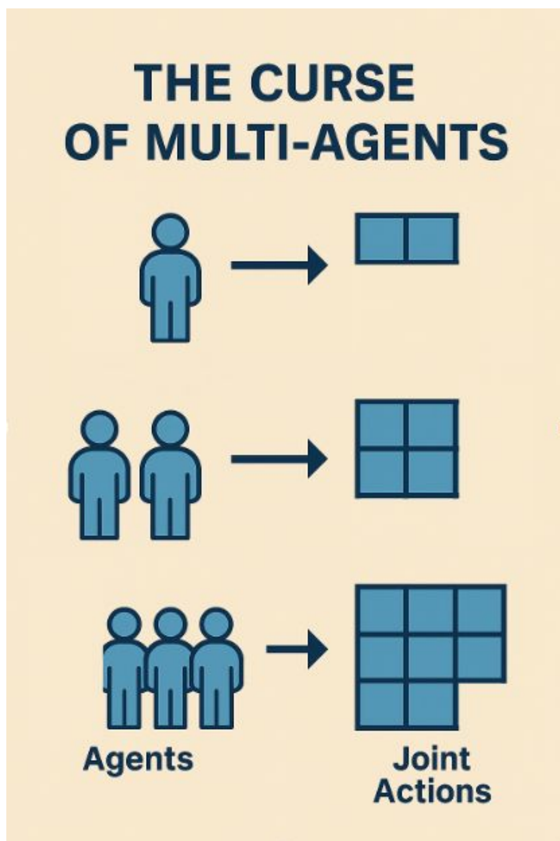
Introduction

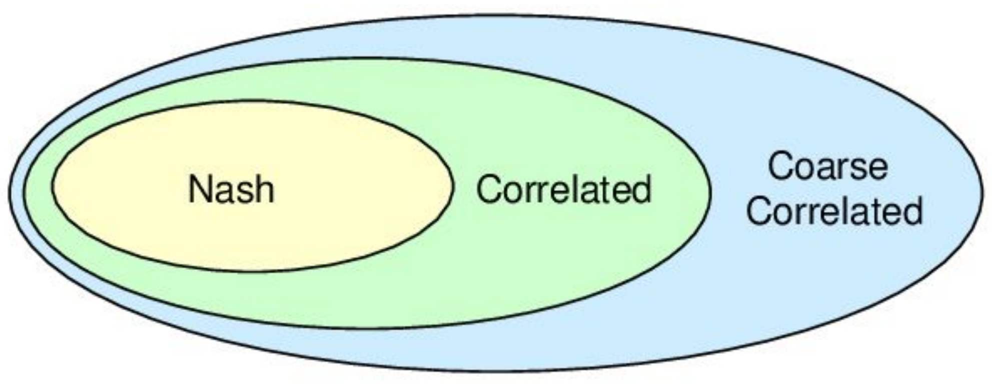
- **What is MARL?**
  Multiple agents learning together—cooperating or competing in a shared environment.
- **Core challenge:**
  Efficient + scalable algorithms
  And convergence to an equilibrium
- **Nash-VI (2-player zero-sum):**
  Optimistic Value Iteration + auxiliary bonus γ
  → efficient model-based.
- **V-Learning (general-sum/decentralized):**
  Per-state adversarial bandits feeding V-updates
  → no joint Q-table needed.
- **MPG-PG (cooperative potential games):**
  Independent policy-gradient updates with provable convergence.
- **All in tabular, episodic Markov games:**
  Strong sample- and iteration-complexity guarantees for each setting.



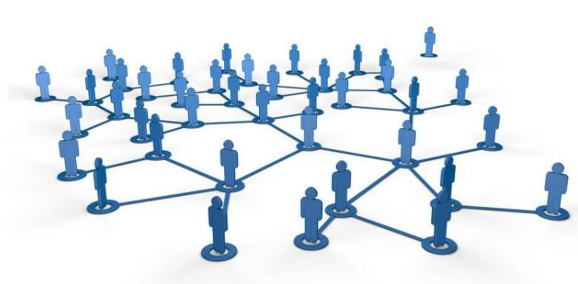
Background







Types of Multi-Agent Markov Games



Sample theoretical lower bounds

Two player zero-sum game	General Markov Game
$\Omega\left(\frac{H^3 S(A+B)}{\epsilon^2}\right)$	$\Omega\left(\frac{H^3 SA}{\epsilon^2}\right)$

Nash VI [1]

Setting: 2-player zero-sum, tabular, finite-horizon, full-observability

Algorithm (per episode):

1. **Optimistic VI**
   - VI with empirical transitions + bonuses ($\beta_t + \gamma_t$)
   - Derive joint policy π^k ← greedy(V)
2. **Rollout & Update**
   - Execute π^k → collect (s,a,s′) → update transition model

$$\begin{aligned}
&\beta \leftarrow \textsc{Bonus}(t, \widehat{\mathbb{V}}_h[(\overline{V}_{h+1} + \underline{V}_{h+1})/2](s,a,b)).\\
&\gamma \leftarrow (c/H)\widehat{\mathbb{P}}_h(\overline{V}_{h+1} - \underline{V}_{h+1})(s,a,b).\\
&\overline{Q}_h(s,a,b) \leftarrow \min\{(r_h + \widehat{\mathbb{P}}_h\overline{V}_{h+1})(s,a,b) + \gamma + \beta, H\}.\\
&\underline{Q}_h(s,a,b) \leftarrow \max\{(r_h + \widehat{\mathbb{P}}_h\underline{V}_{h+1})(s,a,b) - \gamma - \beta, 0\}.
\end{aligned}$$

Key Innovation:

– Auxiliary bonus γ_t captures empirical dynamics (value-gap uncertainty)
– Allows $\beta_t \propto \sqrt{1/t}$ (vs. $\sqrt{S/t}$) → tighter sample complexity

$$\tilde{\mathcal{O}}\left(\frac{H^3 SAB}{\epsilon^2}\right)$$

Main results:

- Improved sample complexity to ε-approximate Nash Equilibrium
- Returns a single (stochastic) Markov policy
- Achieves low regret
- Provides an extension to reward free learning

Limitations:

- Still suffers from the "curse of multi-agents" in more general settings
- Tabular, Finite-Horizon, Full Observability & Known Rewards assumptions
- Computational complexity because Nash-VI requires to solve LPs for computing CCEs in each episode

V-Learning [2]

Goal: Overcome the "curse of multi-agents" via fully decentralized learning—stores only per-state V-values, avoiding exponential state-action blowup.

Approach: Each agent runs its own V-learning loop using adversarial-bandit updates

Complexity: Converges to Nash equilibrium (2-player zero-sum) or ε-approximate CCE (general-sum)

$$\tilde{\mathcal{O}}\left(\frac{H^5 SA}{\epsilon^2}\right)$$

Main results:

- V-learning breaks the "curse of multi-agents"
- Decentralized: No central controller or shared value tables => Scalability
- Plug-and-play: Any adversarial-bandit algorithm with required regret bounds can be used
- Space-efficient variant: Monotonic V-learning reduces memory when agent count is small

Limitations:

- Not optimal with respect to H compared to the theoretical lower bound
- Tabular, Finite-Horizon, Full Observability & Known Rewards assumptions
- Adversarial Bandit Overhead => Each state-step requires running a full bandit algorithm
- Shared Randomness for Correlation => To attain CE or CCE, agents need a common random seed or public signal, which may be nontrivial in fully decentralized systems.

Global Convergence of Multi-Agent Policy Gradient in Markov Potential Games [3]

Setting:
– Cooperative Markov Potential Games (MPGs) with global potential ϕ aligning all agents' incentives
– Direct tabular parameterization

$$V_s^i(\pi) = \Phi_s(\pi) + U_s^i(\pi_{-i})$$

Algorithm:
– Each agent independently performs policy-gradient steps on its own value ∇V_i
– Two variants:

1. **Exact gradient:** full knowledge of dynamics & rewards $\mathrm{O}\left(\frac{1}{\epsilon^2}\right)$
2. **Stochastic gradient:** trajectory sampling → unbiased ∇-estimates $\mathrm{O}\left(\frac{1}{\epsilon^6}\right)$

$$\pi_i^{(t+1)} = \mathrm{Projection}\left(\pi_i^{(t)} + \eta \nabla_{\pi_i} V^i(\pi^{(t)})\right)$$

Main results:

- Definition of Markov Potential Games (MPGs) and providing properties (e.g. existence of Nash Eq.)
- Iteration complexity bounds for achieving an ε-approximate Nash equilibrium
- Fully decentralized => Scalability

Limitations:

- No coverage of general-sum Markov games
- A lot weaker bound for stochastic gradient (which is the most common in practice)
- Potential function requirement => Verifying that a given Markov game admits a global potential can be nontrivial and restrictive; many real-world interactions are neither zero-sum nor fully aligned.
- Assumes direct tabular parametrization
- Tabular, Finite-Horizon, Full Observability & Known Rewards assumptions

Comparative Analysis

Paper	Game Type	Equilibrium	Sample or Iteration Complexity	Centralized or Decentralized	Policy Type
Nash-VI	2-player zero-sum	ε-Nash	$\tilde{\mathcal{O}}\left(\frac{H^3 SAB}{\epsilon^2}\right)$	centralized	single stochastic Markov
V-Learning	General-sum	ε-CE	$\tilde{\mathcal{O}}\left(\frac{H^5 SA^2}{\epsilon^2}\right)$	decentralized	stochastic Markov (mixed)
V-Learning	2-player zero-sum/General-sum	ε-Nash/ ε-CCE	$\tilde{\mathcal{O}}\left(\frac{H^5 SA}{\epsilon^2}\right)$	decentralized	stochastic Markov (mixed)
Multi-Agents Policy Gradient	MPG	ε-Nash	Exact: $\mathrm{O}\left(\frac{1}{\epsilon^2}\right)$ Stochastic: $\mathrm{O}\left(\frac{1}{\epsilon^6}\right)$	decentralized	single stochastic Markov

While each method offers strong theoretical guarantees, they come with trade-offs:

- Nash-VI is highly sample-efficient but limited to two-player zero-sum games and requires centralized control with heavy computations
- V-Learning breaks the curse of multi-agents with a decentralized algorithm, but has suboptimal dependence on the episode length H and in the multi-agents setting finds only ε-approximate CCE or ε-approximate CE.
- Multi-Agents Policy Gradient achieves decentralized learning of ε-approximate Nash equilibrium with multiple agents, but only in MPG, and suffers from slow convergence with stochastic gradients

Conclusion

This project analyzed three theoretical MARL methods: Nash-VI, V-Learning, and MPG Policy Gradient, each tailored to specific game settings and challenges. Nash-VI achieves near-optimal sample efficiency but is restricted to two-player zero-sum games. V-Learning is flexible, decentralized, and scalable to general-sum games, but involves higher complexity and weaker equilibrium guarantees (CE/CCE). Multi-Agents Policy Gradient methods are decentralized, stable, and efficient in Markov Potential Games but rely on strong assumptions and do not directly provide sample-complexity bounds. All those three paper complete the scope of MARL in Markov Games.

A promising future direction is developing MARL algorithms that integrate these strengths—decentralized like V-Learning, efficient in terms of sample complexity like Nash-VI, and broadly applicable with strong equilibrium guarantees as in MPG settings. Further research should also aim to close the gap to theoretical lower bounds, enabling MARL solutions for more complex and realistic scenarios.

References

[1] Qinghua Liu, Tiancheng Yu, Yu Bai, and Chi Jin. A sharp analysis of model-based reinforcement learning with self-play. In International Conference on Machine Learning, pages 7001–7010. PMLR, 2021.
[2] Chi Jin, Qinghua Liu, Yuanhao Wang, and Tiancheng Yu. V-learning–a simple, efficient, decentralized algorithm for multiagent rl. arXiv preprint arXiv:2110.14555, 2021.
[3] Stefanos Leonardos, Will Overman, Ioannis Panageas, and Georgios Piliouras. Global convergence of multi-agent policy gradient in markov potential games. arXiv preprint arXiv:2106.01969, 2021.